Charugundla Anju Bhargavi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.75
Sri Vivekananda Junior College	Class XII (BIEAP)	2024	97.80%
Lakshmi Synergy School	Class X (BSEAP)	2022	96.50%

SCHOLASTIC ACHIEVEMENTS

- Secured 2419 rank in AP EAPCET, in which 0.33M students appeared for the exam, 2024.
- All India Rank 9 in the Unified International English Olympiad (UIEO), 2019
- All India Rank 135 in the Unified International Maths Olympiad (UIMO), 2019
- Secured State Rank 25 in State Level Science Talent Search Examination (SLSTSE), 2017

PROJECTS

An EIS platform (Project report)

Advisor: Prof. Suresh Kumar Garlapati, IIT Hyderabad

- Automated impedance analysis using a function generator and a scope, making EIS more accessible for lab students.
- Extended software to calculate capacitance & permittivity, and plot Bode, Nyquist, & various plots for easier data analysis.
- Tested various ceramic disc capacitors and verified the data with Aftermath software.

MoS2 synthesis and characterization

Advisor: Deepu J Babu, IIT Hyderabad

- Synthesized few-layer MoS via APCVD in a single-zone furnace, optimizing growth at 850°C.
- Characterized MoS using Raman spectroscopy, confirming the 2H semiconducting phase and estimating layer thickness.
- Investigated MoS growth and morphology under varying carrier gases and growth conditions.

3D-Printed working rolling machine:

Advisor: Shourya Dutta Gupta, IIT Hyderabad

- Designed a functional rolling machine in Solid Edge with precise dimensions and detailed 3D models.
- Fabricated the components using 3D printing and assembled them into a working prototype.
- Validated the design through successful operation, demonstrating practical CAD and rapid prototyping skills.

RELEVANT COURSES

Process Metallurgy & Thermal Science: Steelmaking, Ironmaking, Thermodynamics and Kinetics, Phase Transformations, Transport Phenomena.

Process Metallurgy & Thermal Science: Powder Metallurgy, Steelmaking, Ironmaking, Thermodynamics and Kinetics, Phase Transformations, Transport Phenomena.

Mathematical & Computational: Computational Methods in Material Science, Fundamentals of Scientific Computing, Maths for physics, Calculus 1&2, Elementary Linear Algebra, Differential equations

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Physical Metallurgy

- Invigilating examinations and evaluating students' answer scripts.

Multimedia Coordinator

- Hosted an interview and captured multimedia content.

Mentor, Buddy

- Organised stress-free environments for mentor-mentee meetings to enhance connection.
- Volunteered in organising Sunshine events

CERTIFICATES

Summer School on Semiconductor Technology

Centre for Nano Science and Engineering (CeNSE) IISc, Bangalore, India- View Certificate

Certification of Completion of Python Course

Codedex - View Certificate

LAB EXPERIENCE

- **XRD Characterisation**: XRD analysis of BaTiO3 (tetragonal) and Nd2Ti2O7 (monoclinic) are done. Stress analysis is done on crushed, and hammered Al can. (Report available *here*)
- **Metallographic Lab**: A Nickel sample is given to us, which is then prepared for imaging, to see the microstructure of the given sample. Grain size is calculated. (*Reports*)
- **Microstructure & Properties Lab**: Conducted Hall & four-probe measurements, studied dielectric phase transformations of BaTiO3, & analysed polarisation mechanisms in dielectrics. (Reports available *here*)
- Others: Physics Lab, Chemistry Lab

SKILLS

- Technical skills: CAD & 3D printing, Python(Numpy, matplotlib, PyVisa), HTML, CSS.
- Soft skills: Creative, Critical thinking, Curiosity, willingness to learn, and contribute
- Languages: Telugu (Native), English (Fluent), Hindi (Basic), Japanese (Beginner)

EXTRACURRICULAR

- **Social Work**: Volunteered with PRAYAS IIT Hyderabad and National Service Scheme (NSS) by participating in orphanage visits, campus tours for government school students, and campus clean-up drives.
- **Arts**: Enjoy crocheting, as it gives me the satisfaction of recreating things I like. Listening and writing poetry.
- **Sports and games**: Enjoy solving puzzles, playing badminton and carrom in free time.